

## Project Planning Phase

### Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 7 March 2026                                 |
| Team ID       | PNT2022TMIDxxxxxx                            |
| Project Name  | <b>Malaria Detection using Deep Learning</b> |
| Maximum Marks | 8 Marks                                      |

#### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                  | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1 | Registration                  | USN-1             | As a user, I can register for the malaria detection application by entering my email and password. | 2            | High     | Member-1     |
| Sprint-1 | Email Confirmation            | USN-2             | As a user, I receive a confirmation email after registering for the application.                   | 1            | High     | Member-2     |
| Sprint-1 | Login                         | USN-3             | As a user, I can log into the application using my email and password.                             | 1            | High     | Member-1     |
| Sprint-1 | Dashboard                     | USN-4             | As a user, I can access the dashboard after logging into the system.                               | 2            | High     | Member-3     |
| Sprint-2 | Image Upload                  | USN-5             | As a user, I can upload a blood smear image for malaria detection.                                 | 3            | High     | Member-2     |
| Sprint-2 | Image Processing              | USN-6             | As a system, I process the uploaded image before analysis.                                         | 3            | High     | Member-3     |
| Sprint-2 | Detection Model               | USN-7             | As a system, I analyze the image using a machine learning model to detect malaria parasites.       | 5            | High     | Member-1     |
| Sprint-3 | Result Display                | USN-8             | As a user, I can view the malaria detection result (Parasitized or Uninfected).                    | 2            | High     | Member-2     |
| Sprint-3 | Report Generation             | USN-9             | As a user, I can download the malaria detection report.                                            | 2            | Medium   | Member-3     |
| Sprint-3 | History Records               | USN-10            | As a user, I can view previous detection results in the dashboard.                                 | 2            | Medium   | Member-1     |

## References

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3. Kaggle  
Malaria Cell Images Dataset for training machine learning models.  
<https://www.kaggle.com/datasets/iarunava/cell-images-for-detecting-malaria>
4. TensorFlow  
TensorFlow Documentation – Deep learning model development.  
<https://www.tensorflow.org/>
5. OpenCV  
OpenCV Documentation – Image processing techniques.  
<https://opencv.org/>
6. IBM  
IBM Cloud Architecture Center – Technical architecture examples.  
<https://www.ibm.com/cloud/architecture>
7. Amazon Web Services  
AWS Architecture Center – System design and deployment.  
<https://aws.amazon.com/architecture/>
8. C4 Model  
The C4 Model for visualising software architecture.  
<https://c4model.com/>